CHAPTER 6

Molecular aspect of gallstone formation: a systematic review

Ayan Chatterjee[1] and Rukshana Irani[2]
[1]Department of Allied Health, School of Health and Medical Sciences, Adamas University, Kolkata, West Bengal, India
[2]Department of Food & Nutrition, Raidighi College, Radighi, West Bengal, India

Introduction

Gallstones are solid, pebble-like deposits that typically occur in the gallbladder and are composed of bilirubin or cholesterol. They are quite widespread among the population and may or may not cause any symptoms (Lamont et al., 1984). They are one of the most painful biliary tract diseases in adults, with a high incidence, particularly in India (Drury, 1965; Singh et al., 2008). Gallstone disease (GD) is one of the most prevalent diseases in developed countries, but its prevalence is relatively low in developing countries that maintain traditional diets (Schwesinger et al., 1999). Liver cells (hepatocytes) of most vertebrates secrete a green or bitter yellow color fluid called bile, which is stored in the gallbladder and aids in the digestion of fat. Sometimes, the bile stored in the gallbladder hardens into pieces of stone-like materials called gallstones (Cox et al., 2018; Di Ciaula et al., 2018; Reshetnyak, 2012). There are multifactorial causes of GD that include dietary practices, weight loss, gender, and family history, which increase the proneness to the development of such a condition. The chemical makeup of gallstones is influenced by genetics, societal, economic, and dietary variables (Jarrar & Al-Rowaili, 2011). Gallstones can lead to different types of complications as they can block the normal flow of bile during their release from the gallbladder and may get stuck in any of the ducts (hepatic, cystic, and common bile ducts) that carry bile to the intestine from the liver. Trapping or interruption of bile flow in these ducts can lead to inflammation in the gallbladder (cholecystitis), liver, and different other ducts (Cox et al., 2018; Swobodnik et al., 2012).

Compared to males, females experience gallstones twice as frequently. In developed societies, gallstones are a critical problem and affect 10% of the adult population. The prevalence of gallstones in women varies from 5% to 20% for women aged 20–55 years and 25%–30% for women over

Gallstone Formation, Diagnosis, Treatment, and Prevention
DOI: https://doi.org/10.1016/B978-0-443-16098-1.00010-2
© 2024 Elsevier Inc. All rights are reserved, including those for text and data mining, AI training, and similar technologies.

50 years of age (Small, 1980). Gallstones have a close relationship with lipid metabolism. The association between nonalcoholic fatty liver disease and gallstones is bidirectional and complex. Both conditions share common risk factors like overweight, obesity, and the existence of metabolic disorders such as Type II diabetes mellitus, dyslipidemia, and insulin resistance (Brunt et al., 2015).

Prevalence and incidence

According to recent studies, the global scenario of the condition can be portrayed as GD affecting 10%–15% of Europeans, but it affects 3%–5% of people in Asia and Africa. In the American Indian community, female adults can develop GD at a rate of up to 73%. A little over 0.6% of people die each year due to this condition (Ginting, 2011; Marschall & Einarsson, 2007; Stinton & Shaffer, 2012). The Pima tribe of Arizona has the highest gallstone prevalence in the world. A study reported that there were about 70% cases of Pima women who were older than 25 years and had gallstones or a history of cholecystectomy (Lamont et al., 1984).

Pathophysiology of gallstone formation

An abnormal composition of bile acid leads to gallstone formation. Gallstones can be classified into three major types, i.e., pigment stones (black and brown), cholesterol stones, and mixed stones. Pigment stones are composed of calcium bilirubinate, mostly with less than 20% cholesterol content, compared to cholesterol stones containing more than 50% cholesterol monohydrate and also other components that include bile pigment, calcium salt, protein, and fatty acids. The cholesterol content of mixed stones was found to be in between the above two types (Greenberger, 2001; Jaraari et al., 2010). Thickened gallbladder mucoprotein with tiny entrapped cholesterol crystals is called gallbladder sludge and has been thought to be the usual precursor of gallstones (Strasberg et al., 1990). The calcium concentration of bile plays a part in bilirubin precipitation and gallstone calcification (Lopez et al., 1991; Moore, 1990).

Cholesterol gallstones

Cholesterol is the primary component in most of the common types of gallstones. Several mechanisms are involved in their formation, which include monohydrate cholesterol nucleation, supersaturation of bile acid

with cholesterol, and gallbladder hypomotility (Greenberger, 2001; Reshetnyak, 2012). Hypersecretion of cholesterol means an excess amount of bile acids and lecithin than is required. As a result, some of the cholesterol remains undissolved, causing the supersaturation of bile with cholesterol. This may happen due to conditions like obesity, a high-calorie diet, or a high-cholesterol diet. Similarly, if there is hyposecretion of bile acids and lecithin, some of the cholesterol remains undissolved, leading to the same. Thus, in both cases, when the cholesterol level exceeds the level of bile acids and phospholipids, the vesicles rich in cholesterol get aggregated and cholesterol crystals get precipitated. These precipitated, undissolved cholesterol particles stick together and grow in size to eventually form larger gallstones (Reshetnyak, 2012). Another mechanism is by the nucleation of monohydrate cholesterol in bile acid, which may cause an excess of pronucleation factors like mucin and immunoglobulin or a deficiency of anti-nucleation factors such as apolipoprotein AI and AII (Greenberger, 2001) (Figs. 6.1 and 6.2).

Pigment gallstones

These are formed due to the increase of unconjugated bilirubin in the bile (Vítek & Carey, 2012). Calcium salts of bilirubin, phosphate, carbonate, and other anions combine to form pigment gallstones (Singh et al., 2008). When there is excess bilirubin in bile, it combines with other chemical constituents of bile to form pigments. The pigment remains undissolved or gets poorly dissolved in the bile and then sticks together to form particles that slowly grow in size to form hard pigment-type gallstones. Their color may vary from black to brown, depending on their composition. Black pigment stones formed in the gallbladder are irregular and small in size and are made from pure bilirubinate calcium or a polymer-like complex with calcium and glycoprotein mucin. They are generally associated with chronic hemolysis and cirrhosis. Brown pigment

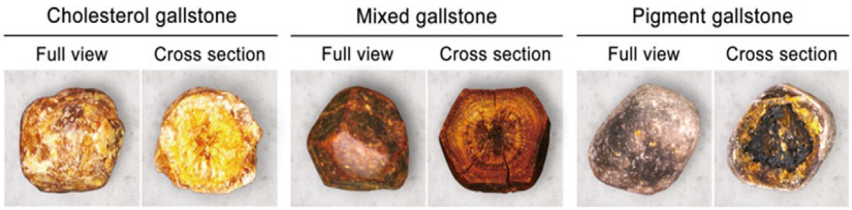


Figure 6.1 A representation of gross and cross-sectional appearances of cholesterol, mixed, and pigmented gallstones (Choi et al., 2019).

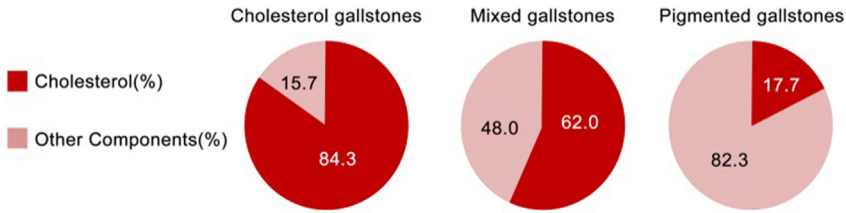


Figure 6.2 The percentage (%) of cholesterol relative to the sum of other components in each type of gallstone (Choi et al., 2019).

stones are made up of unconjugated bilirubin and calcium salt, with variations in the concentration of cholesterol and protein. Brown pigment gallstones are formed in patients with bacterial or parasitic infections and are much softer compared to black pigment gallstones (Cox et al., 2018; Greenberger, 2001; Swobodnik et al., 2012; Van Erpecum, 2011).

Mixed gallstones

These are a mixture of cholesterol and pigment stones that vary in the proportion of bilirubinate salts and cholesterol. Other components of mixed stones include calcium phosphate, calcium carbonate, bilirubin, calcium bilirubinate, calcium stearate, and calcium palmitate (Singh et al., 2008). These stones are sometimes radiographically visible due to the presence of calcium in them (Hess et al., 2011; Singh et al., 2008; Swobodnik et al., 2012).

Molecular mechanisms involved in cholesterol gallstone formation

The variation in composition and biochemical imbalance between the three major lipids present in bile, i.e., cholesterol, bile salts (BSs), and phospholipids, are responsible for the formation of water-insoluble cholesterol crystals. New and more accurate therapeutic strategies can be designed to develop a better understanding of the role of nuclear receptors in the formation of gallstones (Vázquez et al., 2012). Several proteins are responsible for biliary lipid secretion in the liver and are regulated by different transcription factors, including those that belong to the nuclear receptor family, i.e., liver X receptor (LXR) and farnesoid X receptor (FXR), as well as the sterol regulatory element binding proteins (Zanlungo & Rigotti, 2009). Some of the pathological indications of cholesterol GD (CGD) are inflammation of the gallbladder, increased biliary

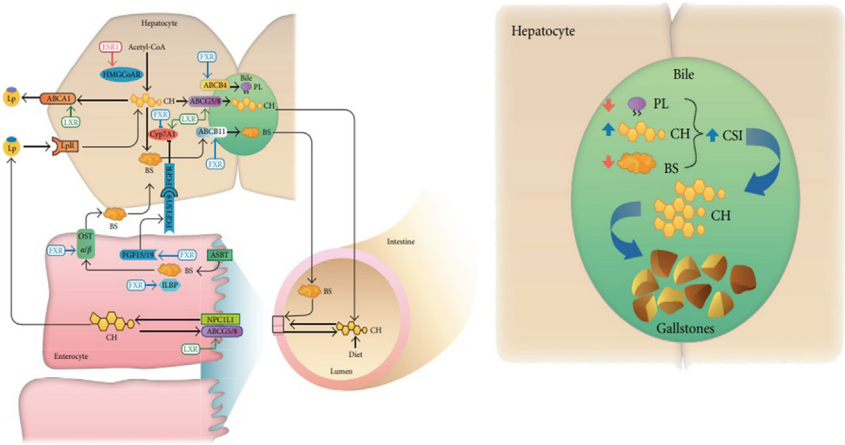
cholesterol secretion, cholesterol microcrystal formation, growth of macroscopic stones due to the aggregation of bile cholesterol in the gallbladder, and also increased bile acid hydrophobicity (Apstein & Carey, 1996; Hofmann, 1998; Portincasa et al., 2006). Cholesterol crystallization due to the imbalance of the lipids is the main factor behind the formation of these stones. Recent studies claim the involvement of some molecular factors behind it. Several biliary proteins, like haptoglobins, immunoglobulins M and G, *aminopeptidase* N, alpha-1 acid glycoprotein, mucin, and alpha-1 anti-chymotrypsin, have been described as nucleating factors that may promote the crystallization of cholesterol. Mucin seems to have a remarkable role in the pathogenesis of CGD in humans (Van Erpecum, 2011).

1. Possible mechanisms of action exhibited by nuclear receptors in the liver and small intestine at the molecular level: Dietary cholesterol as well as the bile enters the intestinal tract, is absorbed by the enterocytes through NPC1L1 (Niemann-Pick C1-like 1), and can be secreted back to the intestinal lumen by ABCG5/G8 (ATP-binding cassette subfamily G, member 5/member 8) transport proteins. Following absorption, cholesterol is incorporated into lipoproteins (Lps) and released into the blood and lymph; it is then transported to the liver after the uptake of triglyceride in the peripheral tissues. BSs are absorbed into the intestine by the apical sodium bile acid transporter (ASBT) and leaked to the basolateral surface by the organic solute transporter alpha/beta (OSTα/β), among others, reaching the liver *via* the systemic circulation. Hepatic cholesterol pools are derived from *de novo* synthesis from acetyl-CoA and selective lipid uptake from receptor-mediated endocytosis and/or Lp. Cholesterol is secreted into plasma high-density lipoprotein *via* adenosine triphosphate binding cassette transporter A1 (ABCA1), by the formation and secretion of very low-density lipoprotein, or is released into the bile through the heterodimeric ABCG5/8 transporter. Bile contains phospholipids (PLs), cholesterol (CH), and BSs. PLs enter the biliary canaliculi *via* the adenosine triphosphate-binding cassette transporter B4 (ABCB4) (a lipid translocator). BSs are derived by the neosynthesis of cholesterol or by uptake from plasma and are then released into the bile by the adenosine triphosphate-binding cassette transporter B11. The secretion and metabolism of lipids are controlled by nuclear receptors (NRs) at different levels. LXR helps to release cholesterol from the intestine and liver through the activation of ABCA1 and ABCG5/8

transporters. Cholesterol 7α−*hydroxylase*/cytochrome P450 7A1 (Cyp7A1) gets activated by LXR, leading to a rise in hepatic BS synthesis. The FXR regulates BS concentration at two different levels, i.e., by promoting the expression of OSTα/β, Fibroblast growth factor 15/19, and ileal lipid binding protein transporters in the intestine, as well as by enhancing the expression of ABCB4 (ATP binding cassette subfamily B, member 4) and ABCC11 (ATP-binding cassette transporter subfamily C, member 11) transporters and repressing the expression of Cyp7A1 in the hepatic tissue. Estrogen receptors increase *de novo* cholesterol synthesis by regulating HMG CoA R (3-Hydroxy-3-methylglutaryl coenzyme A *Reductase)*, ASBT, and the OSTα/β transporter.

2. Cholesterol gallstone formation: An increase in cholesterol levels and/or a lowered BS or PL concentration in the bile leads to an increase in the biliary cholesterol saturation index, which triggers cholesterol precipitation into crystals and thus the formation of cholesterol stones within the gallbladder takes place (Vázquez et al., 2012).

The key characteristics of the mechanism regarding the role of NRs on hepatic and intestinal lipid metabolism and also on transport and CGD have been depicted in Fig. 6.3A and B.



(A) (B)

Figure 6.3 Pathway showing molecular mechanism behind gallstone formation (detailed descriptions are given below).

Role of trace elements in gallstone formation

Recent studies have shown that gallstones not only contain different organic substances but also contain a variety of metal ions such as copper (Cu), zinc (Zn), magnesium (Mg), iron (Fe), calcium (Ca), manganese (Mn), sodium (Na), and aluminum (Al). These components interact with each other. Some factors, like the thermodynamics of bile, an imbalance between promoting and inhibiting factors in bile synthesis, and the kinetics of the gallbladder, are linked with the formation of gallstones. In a recent study, gallstones were analyzed for metal and nonmetal elements. The data implies that sulfur (S), phosphorus (P), Cu, and Ca have a major role in the formation of pigment stones. The higher concentration of Cu and Ca ions was mainly due to the formation of bilirubinates of Cu and Ca. The variation in the composition of pigment stones and mixed stones was much less, as the content of various ions in these stones was quite similar. The existence of proteins in gallstones was expressed by the presence of S and P. Specific proteins, like mucins, promote nucleation in the process of gallstone formation. Cholesterol gallstones have a low metal ion concentration, as their pathogenesis is primarily due to an increase in biliary cholesterol secretion (Zhang et al., 2016).

Genetic mechanism of gallstone formation

Recent studies suggest that genetics play an important role in the formation of gallstones. The risk of formation of gallstones becomes 2−4 times higher if someone in their family is already having it (Abu Eshy et al., 2007; Attili et al., 2005; Zimmer & Lammert, 2011). It is generally characterized by autosomal dominance type of inheritance (Katsika et al., 2005; Qin et al., 2005). Haplotype analysis is a better predictor of risk for GD (Dixit et al., 2007). To better understand the genetic factors involved in gallstone formation, gallstone-resistant mice (AKR/J) and gallstone-susceptible mice (C57L/J) were used to perform an experiment. These mice were used to describe that the lithogenic genes 1 and 2 (Lith1 and Lith2) may play a pivotal role in gallstone formation. Lith1, which is located on mouse chromosome 2, plays a major role in the determination of cholesterol hypersecretion by the liver. Lith2, which is located on mouse chromosome 19, regulates the bile salt-dependent flow of bile (Wang et al., 2018). The functional counterparts of lithogenic gene 1 and lithogenic gene 2 in mice are ABCG5 and ABCG8 in the human

equivalents. ABCG5 and ABCG8 are ATP-binding cassette (ABC) transporters that are significantly expressed in hepatocytes and enterocytes (Wang et al., 2015). The genes associated with the development of GD are assumed to be located mainly on chromosomes 3, 4, 9, and 11 (Qin et al., 2006). Inactivation of ABCG5/G8 transporters will cause a significant reduction in the secretion of cholesterol in bile, which makes the level of cholesterol in the liver and plasma very sensitive to alterations in dietary cholesterol content. As a result, hypercholesterolemia, phytosterolemia, and premature coronary heart disease may occur (Accumulation of Dietary Cholesterol in Sitosterolemia Caused by Mutations in Adjacent ABC Transporters, 2023; Miettinen, 1980). However, the overexpression of ABCG5/G8 proteins augments the cholesterol content in the gallbladder, thus increasing the likelihood of cholesterol crystal precipitation (Identification of a gene, 2023). The carriers of ABCG5 604Q or ABCG8 D19H polymorphisms are more susceptible to the development of GD, irrespective of age, sex, and body mass index (Kuo et al., 2008). The incidence of GD in males has been associated with T400K polymorphisms in ABCG8 proteins (Wang et al., 2007). The increased expression of *3-hydroxy-3-methylglutaryl-coenzyme-A reductase*, the rate-limiting enzyme of cholesterol synthesis, has been earlier suggested to play a major role (Kharitonova, 2003). Genetic polymorphisms in apolipoprotein genes may be associated with an alteration in lipid profile and susceptibility to gallstones (Báez et al., 2010; Jaime et al., 2010). The SNPs (single-nucleotide polymorphisms) at MUC1 (Mucin 1) and MUC2 (Mucin 2) are significantly linked with gallstones in men but not in women. These genes can work together to further increase the susceptibility of the Chinese population to gallstones (Chuang et al., 2011).

Cholecystokinin A receptor gene polymorphism and cholelithiasis

The role of the cholecystokinin A receptor (CCK-AR) in gallstone formation was established by using CCK-AR gene knockout mice. When compared to gallbladders without gallstones, the gene expression of CCK-AR was significantly lower in gallbladders with gallstones. In patients with gallstones, the promoter area had two sequence differences, but no genetic variants were found in the coding region. However, no appreciable variations in the promoter activities of polymorphic promoter constructs were discovered. On the other hand, higher levels of

CCK-AR gene expression were associated with decreased methylation in the promoter region. Gallstone development was accelerated in animals lacking the CCK-AR gene. Alterations in the CCK-AR gene and its polymorphism aid gallstone formation (Miyasaka et al., 2002).

Clinical presentation and diagnosis

Patients often complain about recurrent episodes of right upper quadrant or epigastric pain, probably related to the formation or emergence of stones in the cystic duct (Rigas et al., 1990). A patient may also experience referred pain between the shoulder blades or below the right shoulder region (Boas' sign). Often, pain attacks occur after a particularly heavy, fatty meal and almost always happen at night (Fitzgerald et al., 2009). The three primary methods used to diagnose gallbladder disease are ultrasonography, nuclear scanning (cholescintigraphy), and oral cholecystography. Nowadays, ultrasonography is the method most often used to detect cholelithiasis and cholecystitis. Occasionally, gallstones are also diagnosed during plain X-rays. Ultrasonography has a specificity and sensitivity of 90%–95% and can detect stones as small as 2 mm in diameter (Trotman et al., 1975). A range of different spectroscopic techniques are employed to investigate the chemical composition of gallstones, including those involving X-ray diffraction, elemental analysis, Raman spectroscopy, electron spin resonance spectroscopy, atomic absorption spectroscopy, and the thermogravimetric spectrum (Ferraro et al., 1996; Yang et al., 1992).

References

Abu Eshy, S. A., Mahfouz, A. A., Badr, A., El Gamal, M. N., Al Shehri, M. Y., Salati, M. I., et al. (2007). Prevalence and risk factors of gallstone disease in a high altitude Saudi population. *Eastern Mediterranean Health Journal*, *13*(4), 794–802.

Accumulation of dietary cholesterol in sitosterolemia caused by mutations in adjacent ABC transporters. (2023). Science [Internet]. [cited 2023 May 1]. Available from: https://www.science.org/doi/10.1126/science.290.5497.1771.

Apstein, M. D., & Carey, M. C. (1996). Pathogenesis of cholesterol gallstones: a parsimonious hypothesis. *European Journal of Clinical Investigation*, *26*(5), 343–352.

Attili, A. F., De Santis, A., Attili, F., Roda, E., Festi, D., & Carulli, N. (2005). Prevalence of gallstone disease in first-degree relatives of patients with cholelithiasis. *World Journal of Gastroenterology*, *11*(41), 6508.

Báez, S., Tsuchiya, Y., Calvo, A., Pruyas, M., Nakamura, K., Kiyohara, C., et al. (2010). Genetic variants involved in gallstone formation and capsaicin metabolism, and the risk of gallbladder cancer in Chilean women. *World Journal of Gastroenterology*, *16*(3), 372.

Brunt, E. M., Wong, V. W., Nobili, V., et al. (2015). Nonalcoholic fatty liver disease. *Nature Reviews Disease Primers*, *1*, 15080.

Choi, H. J., Cho, S. J., Kim, O. H., Song, J. S., Hong, H. E., Lee, S. C., et al. (2019). Efficacy and safety of a novel topical agent for gallstone dissolution: 2-methoxy-6-methylpyridine. *Journal of Translational Medicine*, *17*(1), 1–16.

Chuang, S. C., Juo, S. H. H., Hsi, E., Wang, S. N., Tsai, P. C., Yu, M. L., et al. (2011). Multiple mucin genes polymorphisms are associated with gallstone disease in Chinese men. *Clinica Chimica Acta; International Journal of Clinical Chemistry*, *412*(7–8), 599–603.

Cox, M. R., Eslick, G. D., & Padbury, R. (2018). The management of gallstone disease*: A practical and evidence-based approach*. Springer.

Di Ciaula, A., Wang, D. Q. H., & Portincasa, P. (2018). An update on the pathogenesis of cholesterol gallstone disease. *Current Opinion in Gastroenterology*, *34*(2), 71.

Dixit, M., Choudhuri, G., Saxena, R., & Mittal, B. (2007). Association of apolipoprotein A1-C3 gene cluster polymorphisms with gallstone disease. *Canadian Journal of Gastroenterology = Journal Canadien de Gastroenterologie*, *21*(9), 569–575.

Drury, R. A. B. (1965). Gallstones. *Journal of Clinical Pathology*, *18*(4), 484.

Ferraro, J. R., Wu, J. G., Soloway, R. D., Li, W. H., Xu, Y. Z., Xu, D. F., et al. (1996). Copper binding to bilirubin as determined by FT-IR and EPR spectroscopy. *Applied Spectroscopy*, *50*(7), 922–927.

Fitzgerald, J. E. F., Fitzgerald, L. A., Maxwell-Armstrong, C. A., & Brooks, A. J. (2009). Recurrent gallstone ileus: time to change our surgery? *Journal of Digestive Diseases*, *10*(2), 149–151.

Ginting, S. A. (2011). Description characteristic risk factor of the kolelitiasis disease in the Columbia Asia Medan hospital. *Journal Darma Agung*, 38–45.

Greenberger, N. J. (2001). Diseases of the gallbladder and bile ducts. *Harrison's Principles of Internal Medicine*.

Hess, B., Rao, N. P., Preminger, G. M., & Kavanagh, J. P. (2011). Urinary Tract Stone Disease.

Hofmann, A. F. (1998). Progress in idiopathic bile acid malabsorption. *Gut*, *43*(6), 738–739.

Identification of a gene, ABCG5, important in the regulation of dietary cholesterol absorption. (2023). Nature Genetics [Internet]. [cited 2023 May 1]. Available from: https://www.nature.com/articles/ng0101_79.

Jaime, S. C., Maribel, A. M., Eliakym, A. M., José, R. N., Julio, G., Laura, S. M., et al. (2010). ApoB-100, ApoE and CYP7A1 gene polymorphisms in Mexican patients with cholesterol gallstone disease. *World Journal of Gastroenterology*, *16*(37), 4685.

Jaraari, A., Jagannadharao, P., Patil, T., Hai, A., Awamy, H., El Saeity, S., et al. (2010). Quantitative analysis of gallstones in Libyan patients. *Libyan Journal of Medicine*, *5*(1), 4627.

Jarrar, B. M., & Al-Rowaili, M. A. (2011). Chemical composition of gallstones from Al-Jouf province of Saudi Arabia. *The Malaysian Journal of Medical Sciences*, *18*(2), 47.

Katsika, D., Grjibovski, A., Einarsson, C., Lammert, F., Lichtenstein, P., & Marschall, H. U. (2005). Genetic and environmental influences on symptomatic gallstone disease: A Swedish study of 43,141 twin pairs. *Hepatology (Baltimore, Md.)*, *41*(5), 1138–1143.

Kharitonova, L. A. (2003). Cholelithiasis in children: Issues of choice of therapeutic tactics. *Rus Meditsinskiy Zhurnal*, *11*, 787–790.

Kuo, K. K., Shin, S. J., Chen, Z. C., Yang, Y. C., Yang, J. F., & Hsiao, P. J. (2008). Significant association of ABCG5 604Q and ABCG8 D19H polymorphisms with gallstone disease. *British Journal of Surgery*, *95*(8), 1005–1011.

Lamont, J. T., Smith, B. F., & Moore, J. R. (1984). Role of gallbladder mucin in pathophysiology of gallstones. *Hepatology (Baltimore, Md.)*, *4*(S2), 51S–56S.

Lopez, A. J., O'Keefe, P., Morrissey, M., & Pickleman, J. (1991). Ceftriaxone-induced cholelithiasis. *Annals of Internal Medicine*, *115*(9), 712–714.

Marschall, H. U., & Einarsson, C. (2007). Gallstone disease. *Journal of Internal Medicine*, *261*(6), 529–542.

Miettinen, T. A. (1980). Phytosterolaemia, xanthomatosis and premature atherosclerotic arterial disease: A case with high plant sterol absorption, impaired sterol elimination and low cholesterol synthesis. *European Journal of Clinical Investigation*, *10*(1), 27–35.

Miyasaka, K., Takata, Y., & Funakoshi, A. (2002). Association of cholecystokinin A receptor gene polymorphism with cholelithiasis and the molecular mechanisms of this polymorphism. *Journal of Gastroenterology*, *37*, 102–106.

Moore, E. W. (1990). Biliary calcium and gallstone formation. *Hepatology (Baltimore, Md.)*, *12*(3 Pt 2), 206S–214S.

Portincasa, P., Moschetta, A., & Palasciano, G. (2006). Cholesterol gallstone disease. *The Lancet*, *368*(9531), 230–239.

Qin, J., Han, T. Q., Fei, J., Jiang, Z. Y., Zhang, Y., Yang, S. Y., et al. (2005). Risk factors of familial gallstone disease: study of 135 pedigrees. *Zhonghua Yi Xue Za Zhi*, *85*(28), 1966–1969.

Qin, J., Han, T. Q., Yuan, W. T., Fei, J., Jiang, Z. H., Zhang, J., et al. (2006). Study on the position of genes responsible for gallstone disease in Chinese population. *Zhonghua Wai Ke Za Zhi [Chinese Journal of Surgery]*, *44*(7), 485–487.

Reshetnyak, V. I. (2012). Concept of the pathogenesis and treatment of cholelithiasis. *World Journal of Hepatology*, *4*(2), 18–34.

Rigas, B., Torosis, J., McDougall, C. J., Vener, K. J., & Spiro, H. M. (1990). The circadian rhythm of biliary colic. *Journal of Clinical Gastroenterology*, *12*(4), 409–414.

Schwesinger, W. H., Kurtin, W. E., Page, C. P., Stewart, R. M., & Johnson, R. (1999). Soluble dietary fiber protects against cholesterol gallstone formation. *American Journal of Surgery*, *177*(4), 307–310.

Singh, V. K., Singh, V., Rai, A. K., Thakur, S. N., Rai, P. K., & Singh, J. P. (2008). Quantitative analysis of gallstones using laser-induced breakdown spectroscopy. *Applied Optics*, *47*(31), G38–G47.

Small, D. M. (1980). Cholesterol nucleation and growth in gallstone formation. *New England Journal of Medicine. Massachusetts Medical Society*, 1305–1307.

Stinton, L. M., & Shaffer, E. A. (2012). Epidemiology of gallbladder disease: Cholelithiasis and cancer. *Gut Liver*, *6*(2), 172–187.

Strasberg, S. M., Toth, J. L., Gallinger, S., & Harvey, P. R. C. (1990). High protein and total lipid concentration are associated with reduced metastability of bile in an early stage of cholesterol gallstone formation. *Gastroenterology*, *98*(3), 739–746.

Swobodnik, W., Soloway, R. D., & Ditschuneit, H. (2012). *Gallstone disease: Pathophysiology and therapeutic approaches*. Springer Science & Business Media.

Trotman, B. W., Petrella, E. J., Soloway, R. D., Sanchez, H. M., Morris, T. A., III, & Miller, W. T. (1975). Evaluation of radiographic lucency or opaqueness of gallstones as a means of identifying cholesterol or pigment stones: Correlation of lucency or opaqueness with calcium and mineral. *Gastroenterology*, *68*(6), 1563–1566.

Van Erpecum, K. J. (2011). Pathogenesis of cholesterol and pigment gallstones: an update. *Clinics and Research in Hepatology and Gastroenterology*, *35*(4), 281–287.

Vázquez, M. C., Rigotti, A., & Zanlungo, S. (2012). Molecular mechanisms underlying the link between nuclear receptor function and cholesterol gallstone formation. *Journal of Lipids*, *2012*.

Vítek, L., & Carey, M. C. (2012). New pathophysiological concepts underlying pathogenesis of pigment gallstones. *Clinics and Research in Hepatology and Gastroenterology*, *36*(2), 122–129.

Wang, J., Mitsche, M. A., Lütjohann, D., Cohen, J. C., Xie, X. S., & Hobbs, H. H. (2015). Relative roles of ABCG5/ABCG8 in liver and intestine. *Journal of Lipid Research*, *56*(2), 319–330.

Wang, T. Y., Portincasa, P., Liu, M., Tso, P., & Wang, D. Q. H. (2018). Mouse models of gallstone disease. *Current Opinion in Gastroenterology*, *34*(2), 59.

Wang, Y., Jiang, Z. Y., Fei, J., Xin, L., Cai, Q., Jiang, Z. H., et al. (2007). ATP binding cassette G8 T400K polymorphism may affect the risk of gallstone disease among Chinese males. *Clinica Chimica Acta; International Journal of Clinical Chemistry*, *384*(1–2), 80–85.

Yang, Z. H., Wang, K., & Liu, X. T. (1992). Studies of electron spin resonance on bilirubin free radicals. *Science in China. Series B, Chemistry, Life Sciences & Earth Sciences*, *35*(9), 1093–1100.

Zanlungo, S., & Rigotti, A. (2009). Determinants of transhepatic cholesterol flux and their relevance for gallstone formation. *Liver International: Official Journal of the International Association for the Study of the Liver*, *29*(3), 323–330.

Zhang, Y., Zhang, C., Guo, D., Huang, X., Xue, J., & Ouyang, J. (2016). Concentrations of trace elements in different types of gallstones and their effects on gallstone formation. *International conference on biomedical and biological engineering* (pp. 331–336). Atlantis Press.

Zimmer, V., & Lammert, F. (2011). Genetics in liver disease: New concepts. *Current Opinion in Gastroenterology*, *27*(3), 231–239.